



## Reflect with us - Week 8



This assignment will not be graded and is only for practice.

1 point

Choose the set of correct options.

☐

Suppose  $\beta = \{v_1, v_2, \dots, v_n\}$  is an orthogonal basis of an inner product space  $V$ . If there exists some  $v \in V$ , such that  $\langle v, v_i \rangle = 0$  for all  $i = 1, 2, \dots, n$ , then  $v = 0$ .

☐

There exists an orthonormal basis for  $\mathbb{R}^n$  with the standard inner product.

☐

If  $P_W$  denotes the linear transformation which projects the vectors of an inner product space  $V$  to a subspace  $W$  of  $V$ , then  $\text{range}(P_W) \cap \text{null space}(P_W) = \{0\}$ , where  $0$  denotes the zero vector of  $V$ .

☐

$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$  cannot represent a matrix corresponding to some projection.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Suppose  $\beta = \{v_1, v_2, \dots, v_n\}$  is an orthogonal basis of an inner product space  $V$ . If there exists some  $v \in V$ , such that  $\langle v, v_i \rangle = 0$  for all  $i = 1, 2, \dots, n$ , then  $v = 0$ .

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**Solution:**

Option 1: Option 1:

$\beta = \{v_1, v_2, \dots, v_n\}$  is an orthonormal basis of the inner product space  $V$ .

Step 1: As  $v \in V$ , we can express  $v$  as a linear combination of the vectors

$v_1, v_2, \dots, v_n$  as follows:

$v = \alpha_1 v_1 + \dots + \alpha_n v_n$ , where  $\alpha_1, \dots, \alpha_n \in \mathbb{R}$

Taking inner product of  $v$  with  $v_1$ , we have  $\langle v, v_1 \rangle = \langle \alpha_1 v_1 + \dots + \alpha_n v_n, v_1 \rangle = \alpha_1 \langle v_1, v_1 \rangle + \dots + \alpha_n \langle v_n, v_1 \rangle$

Recall two properties of inner products on  $V$ :

i)  $\langle u_1 + u_2, u_3 \rangle = \langle u_1, u_3 \rangle + \langle u_2, u_3 \rangle$  for any vectors

$u_1, u_2, u_3 \in V$ .

ii)  $\langle cu_1, u_2 \rangle = c \langle u_1, u_2 \rangle$  for any vectors  $u_1, u_2 \in V$  and  $c \in \mathbb{R}$ .

Using these properties we have,



$$\langle v, v_1 \rangle = \alpha_1 \langle v_1, v_1 \rangle + \alpha_2 \langle v_2, v_1 \rangle + \dots + \alpha_n \langle v_n, v_1 \rangle \quad \langle v, v_1 \rangle = \alpha_1 \langle v_1, v_1 \rangle + \alpha_2 \langle v_2, v_1 \rangle + \dots + \alpha_n \langle v_n, v_1 \rangle.$$

What is the value of  $\langle v_1, v_1 \rangle \langle v_1, v_1 \rangle$ ?

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 1

1 point

What is the value of  $\langle v_2, v_1 \rangle \langle v_2, v_1 \rangle$ ?

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 0

1 point

What is the value of  $\langle v_i, v_1 \rangle \langle v_i, v_1 \rangle$ , for any  $i \in \{2, 3, \dots, n\}$ ?

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 0

1 point

Feedback: Recall the definition of an orthonormal basis and also observe that  $\beta$  is an orthonormal basis.

If  $\langle v, v_1 \rangle = 0$  and  $\langle v, v_1 \rangle = 0$ , then what is the value of  $\alpha_1$ ?

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 0

1 point

Feedback: Observe that  $\langle v, v_1 \rangle = \alpha_1 \langle v, v_1 \rangle = \alpha_1$

Similarly we can conclude that,  $\alpha_i = 0$ , for each  $i = 1, 2, \dots, n$

Hence,  $v = 0v = 0$ .

Option 2: Let  $\gamma = \{v_1, v_2, \dots, v_n\}$  be a given ordered basis of a vector space. The steps in the process are described below to yield an orthonormal basis  $\beta = \{u_1, u_2, \dots, u_n\}$ .

Step-1  $w_1 = v_1$ , and  $u_1 = \frac{w_1}{\|w_1\|}$

Step-2  $w_2 = v_2 - \langle v_2, u_1 \rangle u_1$  and  $u_2 = \frac{w_2}{\|w_2\|}$

Step-3  $w_3 = v_3 - \langle v_3, u_1 \rangle u_1 - \langle v_3, u_2 \rangle u_2$  and  $u_3 = \frac{w_3}{\|w_3\|}$

Step-i  $w_i = v_i - \sum_{j=1}^{i-1} \langle v_i, u_j \rangle u_j$  and  $u_i = \frac{w_i}{\|w_i\|}$ .

Step-n  $w_n = v_n - \sum_{j=1}^{n-1} \langle v_n, u_j \rangle u_j$  and  $u_n = \frac{w_n}{\|w_n\|}$ .

Hence, Option 2 is correct.

Option 3: The projection of a vector to a subspace: Let  $V$  be an inner product space,  $v \in V$  and  $W \subseteq V$  be a subspace. Then the projection of  $v$  onto  $W$  is the vector in  $W$ , denoted by  $P_W(v)$ , computed as follows:

Find an orthonormal basis  $\{v_1, v_2, \dots, v_n\}$  for  $W$ . Find an orthonormal basis  $\{v_1, v_2, \dots, v_n\}$  for  $W$ .

Then  $P_W(v) = \sum_{i=1}^n \langle v, v_i \rangle v_i$

Step 1:

$$\begin{aligned} P_W^2(v) &= P_W(P_W(v)) \\ &= P_W\left(\sum_{i=1}^n \langle v, v_i \rangle v_i\right) \\ &= \sum_{i=1}^n P_W(\langle v, v_i \rangle v_i) \\ &= \sum_{i=1}^n \langle v, v_i \rangle P_W(v_i) \end{aligned}$$

$$(\langle v, v_i \rangle v_i) = \sum_{i=1}^n \langle v, v_i \rangle P_W(v_i)$$

1 point

At this point can you identify  $P_W(v_i)$ ?

☐

$v_i$

☐ 0

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$v_i$

Feedback: As  $v_i$  is itself in  $W$ , the projection of it onto  $W$  is nothing but the vector  $v_i$  itself. On applying the definition, we get,  $P_W(v_i) = \sum_{j=1}^n \langle v_i, v_j \rangle v_j = v_i$  as  $\langle v_i, v_j \rangle = \delta_{ij}$  where  $\delta_{ij} = 1$  if  $i = j$  and  $\delta_{ij} = 0$  if  $i \neq j$ . Hence we have

Hence we have

$$P_W^2(v) = \sum_{i=1}^n \langle v, v_i \rangle P_W(v_i) = \sum_{i=1}^n \langle v, v_i \rangle v_i = P_W(v)$$

So we can conclude  $P_W^2 = P_W$ .

Step 2: Let  $v \in \text{range}(P_W) \cap \text{null space}(P_W)$ . So  $v \in \text{range}(P_W)$  and  $v \in \text{null space}(P_W)$ .

As  $v \in \text{range}(P_W)$  there exists  $v' \in V$  such that  $P_W(v') = v$ , and as  $v \in \text{null space}(P_W)$ , we have  $P_W(v) = 0$ .

$$\begin{aligned}
 P_W(v') &= v \\
 P_W^2(v') &= P_W(P_W(v')) \\
 P_W(v') &= P_W(v) \text{ [as, } P_W^2 = P_W \text{]} \\
 P_W(v') &= 0 \\
 v &= 0
 \end{aligned}$$

$$(v')) = P_W(v) \text{ [as, } P_W^2 = P_W \text{]} = 0 = 0$$

Hence  $\text{range}(P_W) \cap \text{null space}(P_W) = \{0\}$   $\text{range}(P_W) \cap \text{null space}(P_W) = \{0\}$ .

Option 4: Option 4: If a matrix  $A$  represents a projection, then  $A^2$  must be  $A$ .

Let  $A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$   $A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ .

1 point

Is  $A^2 = AA = A$ ?

☐ Yes

☐ No

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

No

Hence,  $\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$  cannot represent a matrix corresponding to some projection.

[Check Answers](#)

Your score is: 0/7